

MedVoice AI

Component Inventory & Progress Report

Product Engineering

June 2026 — Matrix v1.2

1 Overall Status

Metric	Value
Stage	Prototype
Product-ready	No
Overall completion	~33% toward production medical receptionist
Primary blocker	Phone unverified; HIPAA not implemented; no production deploy

2 Runtime Surfaces

3 Component Registry

3.1 A. Voice Widget Frontend (js/, index.html)

Component	Purpose	Status	%
app.js	UI boot, calendar hooks, health poll	FUNCTIONAL	90
response-orchestrator.js	6-step conversation pipeline	FUNCTIONAL	85
state-machine.js	FSM + engagement scoring	FUNCTIONAL	90
speech-io.js	Browser Web Speech API	FUNCTIONAL	70
config/loader.js	Multi-tenant ?client=	FUNCTIONAL	95
config/validator.js	Schema validation	VERIFIED	100
modules/* (12 files)	Intent, lead, emergency, flows	FUNCTIONAL	80
nlp/* (10 files)	Temporal, confidence, semantic	FUNCTIONAL	75
router/*	Conversation routing	FUNCTIONAL	70
knowledge-engine.js	Config-driven insights	FUNCTIONAL	65
llm-adapter.js	Ollama proxy client	FUNCTIONAL	80
webhook-dispatcher.js	HMAC outbox (browser)	FUNCTIONAL	75
outbox-db.js	IndexedDB queue	FUNCTIONAL	80
local-model-runner.js	WebLLM scaffold	∂	10
Subtotal			78

Improvements needed: Server-side webhook relay; retire browser STT for production phone; add business-hours gate.

3.2 B. Express Backend (server/)

Route / Module	Purpose	Status	%
index.js	App shell, CORS, rate limit	FUNCTIONAL	90
lib/database.js	SQLite WAL, prepared stmts	FUNCTIONAL	85
routes/config.js	Tenant config + secret inject	VERIFIED	95
routes/llm-proxy.js	Ollama proxy + sanitization	FUNCTIONAL	85
routes/log.js	Turn persistence	FUNCTIONAL	80
routes/sessions.js	Session CRUD + turns	FUNCTIONAL	85
routes/leads.js	Lead capture API	FUNCTIONAL	85
routes/emergency.js	Emergency event log	VERIFIED	90
routes/calendar.js	freebusy + events.insert	∂	60
routes/voice.js	Twilio TwiML scaffold	∂	15
routes/vapi.js	Vapi webhooks	NOT STARTED	0
routes/billing.js	Stripe subscriptions	NOT STARTED	0
routes/tenants.js	Tenant CRUD API	NOT STARTED	0
middleware/auth.js	Dashboard JWT auth	NOT STARTED	0
Subtotal			58

Improvements needed: Vapi routes; calendar E2E wiring; server webhook worker; auth middleware.

3.3 C. Admin Dashboard (dashboard/src/)

Panel	Purpose	Status	%
Overview	KPIs, session summary	FUNCTIONAL	70
Leads	Lead table + detail	FUNCTIONAL	75
Sessions	Transcript viewer	FUNCTIONAL	70
Emergency	Red-flag event log	FUNCTIONAL	80
System Health	LLM + server status	FUNCTIONAL	75
Configuration	Read-only tenant config	∂	40
Tenant Management	Multi-clinic admin	NOT STARTED	0
Analytics	Call volume, conversion	NOT STARTED	0
Billing	Usage + invoices	NOT STARTED	0
Auth / Login	RBAC	NOT STARTED	0
Call Recordings	Audio playback	NOT STARTED	0
Subtotal			52

3.4 D. Data & Config

Asset	Purpose	Status	%
configs/medical-clinic.json	Primary tenant config	FUNCTIONAL	90
configs/default.json	Fallback tenant	FUNCTIONAL	80
server/data/medvoice.db	SQLite persistence	FUNCTIONAL	80
Tenant DB schema	Per-clinic credentials store	NOT STARTED	0
Patient records DB	Returning patient lookup	NOT STARTED	0

3.5 E. Tests & DevOps

Item	Coverage	Status	%
Vitest unit tests (4 files)	Config, emergency, lead, temporal	FUNCTIONAL	40
E2E booking test	Calendar flow	NOT STARTED	0
CI/CD (GitHub Actions)	Build + test pipeline	NOT STARTED	0
Production deployment	Hosted URL	NOT STARTED	0
Monitoring (Sentry/Datadog)	Error + uptime	NOT STARTED	0
HIPAA hosting + BAA	Compliance layer	NOT STARTED	0

4 Missing Components (Must Build)

1. **Vapi integration layer** — assistant-request, tool-calls, end-of-call-report
2. **Server-side webhook worker** — replace browser-only delivery for phone
3. **Tenant management API + UI** — onboard clinics without editing JSON
4. **Authentication / RBAC** — protect dashboard and admin APIs
5. **Billing (Stripe)** — subscription tiers, usage metering
6. **HIPAA stack** — encrypted PHI store, audit logs, BAAs, HIPAA-eligible hosting
7. **Call transfer / escalation** — Twilio transfer from Vapi tool
8. **SMS confirmations** — Twilio SMS post-booking
9. **After-hours logic** — `business_hours` in config + flow branching
10. **Reschedule / cancel** — calendar event update/delete tools

5 Capability Matrix Summary

Category	Done	Scaffold	Missing	Total
Core receptionist	7	2	8	17
Platform / ops	3	1	8	12
Total	10	3	16	29

6 Per-Component Feature Roadmap

Component	Next Features
Orchestrator	Deprecate for phone; keep for web demo widget
Calendar	Enable <code>calendar_enabled</code> ; timezone handling; buffer slots
Leads	CRM sync (HubSpot/Salesforce webhook templates)
Emergency	Auto-transfer to on-call number via Vapi
Dashboard	Editable config, call playback, per-tenant views
Multi-tenant	Phone-number → tenant mapping table
Webhooks	Server queue + retry + delivery dashboard
LLM	Production model (GPT-4o mini); prompt versioning per tenant